

Extending the GLM: Nominal, Ordinal, and Count Outcomes

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Chapter overview

An outcome's measurement determines what a model must represent. This chapter extends regression to three kinds of outcome: a category with no assumed ordering, an ordered category with unknown spacing, and a count observed over a specified window. A household employment study, four countries' opinions and school absences connect the equations to the comparisons they answer. The chapter develops probability compositions, cumulative odds, exposure offsets and conditional dispersion, and uses fitted values to make the size of each comparison intelligible.

Three questions motivate the models. Among 263 Canadian women surveyed in 1977, how does labour-force participation differ across household income and the presence of children? Among 5,381 respondents in four countries, how does opinion about government effort for people in poverty differ across country and recorded characteristics? Among 146 Australian pupils, how does expected absence in an academic year differ across the study's recorded groups? The outcomes are respectively a composition, an ordered response and a count. Each needs a mean structure respecting what was measured, rather than an arbitrary numerical score followed by a line.

The records come from `carData::Womenlf`, `carData::WVS` and `MASS::quine` (Fox et al., 2022; Fox & Weisberg, 2019; Venables & Ripley, 2002). The package extracts carry no sampling weights, clustering or stratification information. Their availability does not establish a probability sample, so the inferential readings below are conditional comparisons among the recorded units under working independent-row models. The exposure simulation later is explicitly invented. The distinction between those sources matters for what a model can demonstrate and what a study can establish.

Displayed results are rounded to four decimal places, with trailing zeros dropped unless they align a comparison. Calculations using different rounding or software settings may differ slightly.

1 One Family, Three Outcomes

1.1 Keep the predictor; change the response structure

A generalized linear model separates a response distribution, a linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, and a link connecting the conditional mean to that predictor (Nelder & Wedderburn, 1972). A binary logistic model uses a Bernoulli outcome and a logit link. Nominal and ordinal models extend the idea to several category probabilities; count models connect a nonnegative expected count to a log predictor. These are related regression structures, not identical single-response GLMs in every detail. In the vector notation, $\mathbf{x}_i$ includes an initial 1 when the model contains an intercept. The ordered model below uses thresholds instead and omits that entry.

Table 1

A common predictor framework, with different response and variance specifications. The linear predictor is the same object throughout.

Outcome	Random component	Link and mean	Conditional variance	Home
Quantitative score	Normal	identity, $\mu_i = \eta_i$	σ^2 , a free parameter	Ch. 7–9
Binary event	Bernoulli	logit, $\pi_i = \text{logit}^{-1}(\eta_i)$	$\pi_i(1 - \pi_i)$	Ch. 11
Unordered categories	Categorical, K classes	generalized logit, $K - 1$ equations	multinomial	Sec. 2
Ordered categories	Categorical, K ordered classes	cumulative logit, $K - 1$ thresholds	multinomial	Sec. 3
Count or rate	Poisson	log, $\mu_i = t_i e^{\eta_i}$	μ_i	Sec. 4
Overdispersed count	Quasi-Poisson or negative binomial	log, as above	$\phi\mu_i$ or $\mu_i + \mu_i^2/\theta$	Sec. 5

The predictor remains a sum of terms, but a coefficient’s interpretation changes with the response and link. Full probability models use maximum likelihood. Quasi-Poisson is the exception: it specifies a mean and variance without a full probability distribution, so it has no ordinary likelihood or information criterion.

1.2 Decide what the outcome records

Table 2

Outcome triage. The classification is a measurement judgment made before any model is fitted.

Outcome kind	What one value records	Model in this collection
Binary event	Whether an event occurred	Logistic regression, Chapter 11
Nominal categories	A category with no defensible ordering: field of study, party voted for, mode of transport	Generalized (baseline-category) logits, Section 2
Ordered categories	A rank on one substantive dimension with unknown spacing: agreement, severity stage, service rating	Cumulative logits, Section 3
Count	How many events occurred in a window supplied by the design	Poisson-family regression, Sections 4–5
Rate	How many events per unit of exposure, when windows differ across units	Count regression with an exposure offset, Section 4.3
Proportion of known trials	How many of a known number of trials succeeded	Binomial regression with a denominator, named only
Duration until an event	How long until an event, possibly not yet observed	Survival models, outside this collection

The distinction between nominal and ordered categories is substantive. Employment status can be studied as different roles, or ordered by work intensity if that is the declared construct. Coding an order does not establish it. Even when an order is defensible, the distances between categories need not be known. Giving them scores 0, 1 and 2 imposes both ordering and equal spacing; it answers a mean-score question that may not match the intended composition question.

```
dat$score <- ifelse(dat$partic == "not.work", 0,
  ifelse(dat$partic == "parttime", 1, 2))
scored <- lm(score ~ hincome + children, data = dat)
range(fitted(scored))
```

The scored labour-force fit ranges from -0.2099 to 1.5278 , including four negative fitted means, and supplies no probability for any category. A count likewise needs its observation window: ten incidents in one month and ten in a year are different rates. A successes count out of a known number of trials can instead be modeled as binomial; having a count does not automatically imply a Poisson response.

All three empirical examples use the full distributed extract, with no missing analysis values. `Womenlf` records 155 not working, 66 full time and 42 part time. `WVS` has 2,708 Too Little,

1,862 About Right and 811 Too Much responses. `quine` has integer absence counts from 0 to 81, including nine zeros. Auditing those definitions and denominators comes before fitting.

```
table(dat$partic)
table(opinion$poverty)
range(absence$Days)
colSums(is.na(opinion))
```

1.3 Likelihood and comparison carry over

A likelihood evaluates the observed outcomes under specified probabilities, with the data fixed. Maximizing it selects coefficients; the **deviance** compares that maximum with a saturated model fitting each observation:

$$D = 2\{\ell_{\text{saturated}} - \ell(\widehat{\beta})\}. \quad (1)$$

Differences between regular nested models with the same outcome and rows can yield likelihood-ratio tests. AIC and BIC add a parameter penalty to -2ℓ . A smaller deviance by itself is neither proof of adequate fit nor permission to compare any two models by chi-square. In particular, changing the family can change the saturated likelihood, and quasi-Poisson has no full likelihood. Section 6 makes these distinctions concrete.

2 Nominal Outcomes: The Generalized Logit

2.1 Several category probabilities, one reference

Let Y_i take one of K unordered categories, with $\pi_{ik} = P(Y_i = k \mid \mathbf{x}_i)$. Choosing category K as the baseline, the **generalized logit**, or baseline-category multinomial logit, states

$$\log \frac{\pi_{ik}}{\pi_{iK}} = \beta_{0k} + \sum_{j=1}^p \beta_{jk} x_{ij}, \quad k = 1, \dots, K-1. \quad (2)$$

There are $K-1$ equations, each with its own intercept and slopes. One baseline sets the coordinate system; it does not make the category more fundamental. Writing its predictor as $\eta_{iK} = 0$, the probabilities are

$$\pi_{ik} = \frac{e^{\eta_{ik}}}{\sum_{m=1}^K e^{\eta_{im}}}. \quad (3)$$

They sum to one by construction. A new reference changes the printed coefficient equations while preserving the same fitted probabilities. The rows must still support the comparisons: a tiny or empty category within a predictor region can destabilize an estimate.

2.2 Name both categories in every ratio

The exponentiated coefficient $e^{\beta_{jk}}$ is the factor by which π_k/π_K differs for units one predictor unit apart, holding the other predictors fixed. It is conventionally called a **relative-risk ratio** (RRR), also interpretable as a pairwise-category odds ratio when the comparison is restricted to those two categories. It is not the cumulative odds ratio developed in Section 3.

An RRR above one need not make category k 's own probability rise: a third category may rise faster, changing the shared denominator. The ratio names both compared categories, its predictor unit and adjustment set. Probabilities then show the whole composition, which a single pairwise ratio cannot do (Long & Ervin, 2000).

2.3 Labour-force composition in the household records

The unit is one recorded woman. Husband's income is in thousands of 1977 dollars, and children are coded absent or present. We ask about conditional labour-force composition within these records, choosing not working outside the home as the baseline. With children absent, 46 of 79 women worked full time; with children present, 20 of 184 did. Part-time counts are 7 and 35. These observations motivate separate category probabilities, without themselves proving or disproving an ordered model.

```
dat$partic <- relevel(dat$partic, "not.work")
nom <- nnet::multinom(partic ~ hincome + children, data = dat, trace = FALSE)
exp(coef(nom))
exp(confint(nom))
```

Table 3

Generalized logits for labour-force participation, with not working outside the home as baseline. Deviance 422.8819 on six parameters; AIC 434.8819.

Equation and term	Coefficient	SE	RRR	95% interval
Full time: intercept	1.9828	0.4842	—	—
Full time: husband's income	−0.0972	0.0281	0.9073	[0.8587, 0.9587]
Full time: children present	−2.5586	0.3622	0.0774	[0.0381, 0.1574]
Part time: intercept	−1.4323	0.5925	—	—
Part time: husband's income	0.0069	0.0235	1.0069	[0.9617, 1.0543]
Part time: children present	0.0215	0.4690	1.0217	[0.4075, 2.5619]

An additional thousand dollars of husband's income multiplies the full-time-to-not-working probability ratio by .9073. Children present are associated with .0774 times that ratio, at the same income. The part-time-to-not-working intervals both cover one. That leaves a wide range of compatible comparisons rather than proving no association. Each predictor appears in two equations, so testing its removal as a block uses two df: the income block gives $\chi^2(2) = 15.1532$, $p = .0005$, and the children block 63.5586, $p < .0001$.

2.4 Read the fitted composition

```
grid <- expand.grid(hincome = c(5, 15, 25),
  children = c("absent", "present"))
predict(nom, grid, type = "probs")
```



Figure 1: Fitted participation probabilities across income, with observed income support shown below.

Full-time probabilities decline with income, while part-time probabilities rise. The middle-category rise is possible under a proportional-odds model too; it is not by itself evidence against ordering. The curves answer the composition question directly: each vertical slice allocates one whole probability across three categories. With children absent, full-time probability falls from .7817 at income 5 to .3323 at 25; with children present it falls from .2164 to .0369. Thus the

same income relative-risk ratio need not imply the same probability difference. Predict how changing the reference category would change this figure: its fitted curves would stay in place. At income 25 with children present the observed full-time cell is empty, making the small fitted probability dependent on the regression's pooling across income values, rather than a precise direct estimate for that cell.

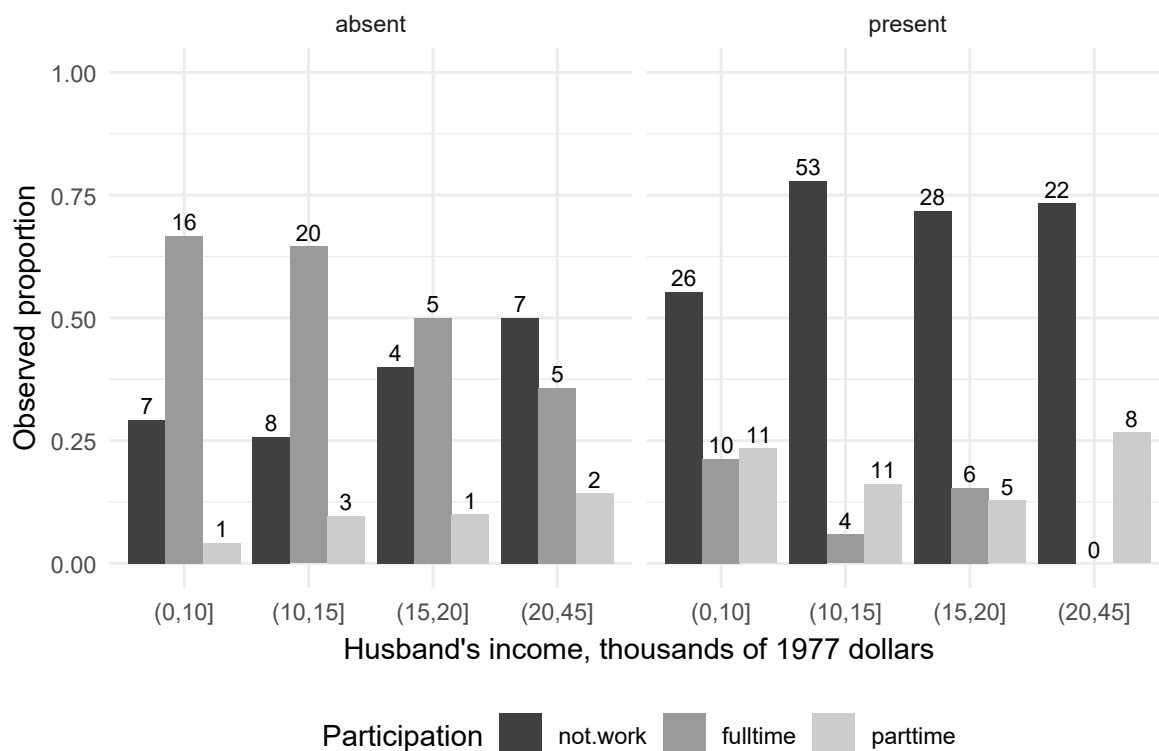


Figure 2: Observed participation composition by income band and children present; bar labels are counts.

2.5 A different reference gives the same fitted surface

With full-time work as the reference, the not-working coefficients become -1.9828 , $.0972$ and 2.5586 ; the part-time coefficients become -3.4151 , $.1041$ and 2.5801 . Deviance remains 422.8819. Fitted probabilities agree with the first fit to within 5×10^{-5} , reflecting numerical tolerance. Predict this invariance before refitting: relabeling a denominator cannot change the modeled distribution of the same outcome.

```

other <- dat
other$partic <- releval(other$partic, "fulltime")
refit <- nnet::multinom(partic ~ hincome + children, data = other, trace = FALSE)
predict(refit, grid, type = "probs")

```

These are cross-sectional comparisons. “Women whose husbands earned more were less likely to work full time” describes them; “raising the husband’s income would move a woman out of full-time work” requires a causal design and assumptions the records do not supply.

3 Ordered Outcomes: The Cumulative Logit

3.1 Cumulative probabilities and thresholds

For a defensible order $1 < \dots < K$, define $\gamma_{ij} = P(Y_i \leq j \mid \mathbf{x}_i)$ for $j = 1, \dots, K - 1$. Each is a cumulative probability at one **cut point**. Category probabilities are differences of successive cumulative probabilities. A cumulative logit model with proportional odds is

$$\log \frac{P(Y_i \leq j \mid \mathbf{x}_i)}{P(Y_i > j \mid \mathbf{x}_i)} = \zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \zeta_1 < \dots < \zeta_{K-1}. \quad (4)$$

The ordered ζ_j are thresholds, not effects. The predictor vector here omits the intercept entry because the thresholds supply the location constants. The minus sign used by `ordinal::clm()` and `MASS::polr()` means that a positive coefficient shifts probability toward higher categories. Thus e^{β_j} multiplies the odds **above** either cut, not a single pair of categories.

One useful picture imagines a continuous response $Y_i^* = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with standard logistic error, observed only as the interval between fixed thresholds into which it falls. The shift changes how much probability lies in each interval. This is a representation of the model, not proof that an actual latent attitude has a logistic distribution.

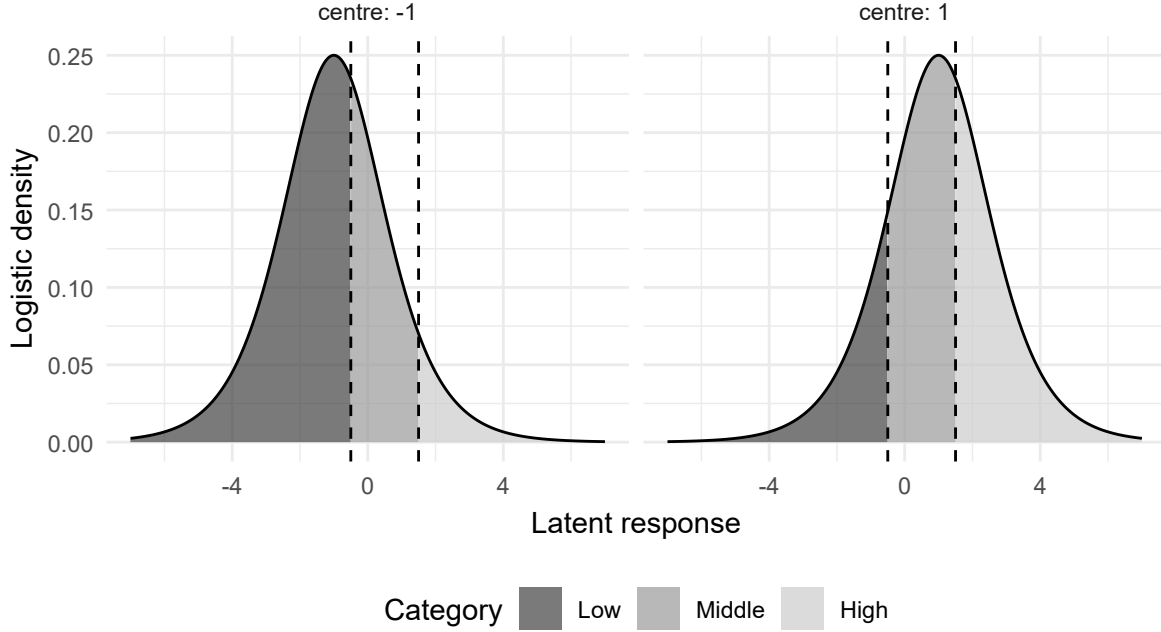


Figure 3: The same thresholds divide two shifted logistic distributions into three ordered categories.

3.2 The common-slope restriction is testable

The model uses the same slope vector at every cut. Consequently the cumulative log-odds difference between two otherwise comparable units is the same at each cut: **proportional odds**. A cut-specific alternative replaces β by β_j . The common-slope model is nested in that alternative, so their likelihood comparison tests a real restriction (McCullagh, 1980). Ordering the outcome alone does not impose or justify it.

Start with a process whose truth is known. Imagine a study of confidence, recorded as Low, Middle or High, and a preparation score x rescaled to lie between -1 and 1 . We generate independent records with x uniform over that range. For the two cumulative probabilities, set

$$\text{logit}\{P(Y \leq 1 | x)\} = -.5 - b_1 x, \quad \text{logit}\{P(Y \leq 2 | x)\} = 2 - b_2 x. \quad (5)$$

These cumulative probabilities stay ordered throughout the chosen range, so their differences give valid Low, Middle and High probabilities. The first process uses $b_1 = b_2 = .6$: preparation has the same above-cut odds ratio at both splits. The second uses $b_1 = .3$ and $b_2 = 1.5$:

preparation is much more strongly related to reaching High than to leaving Low. Both processes describe ordered responses; only the first has proportional odds.

Predict and explain. We fit a common-slope model to 4,000 records from each process. Which fitted lines must be parallel, and where might that restriction conceal a category-probability difference?

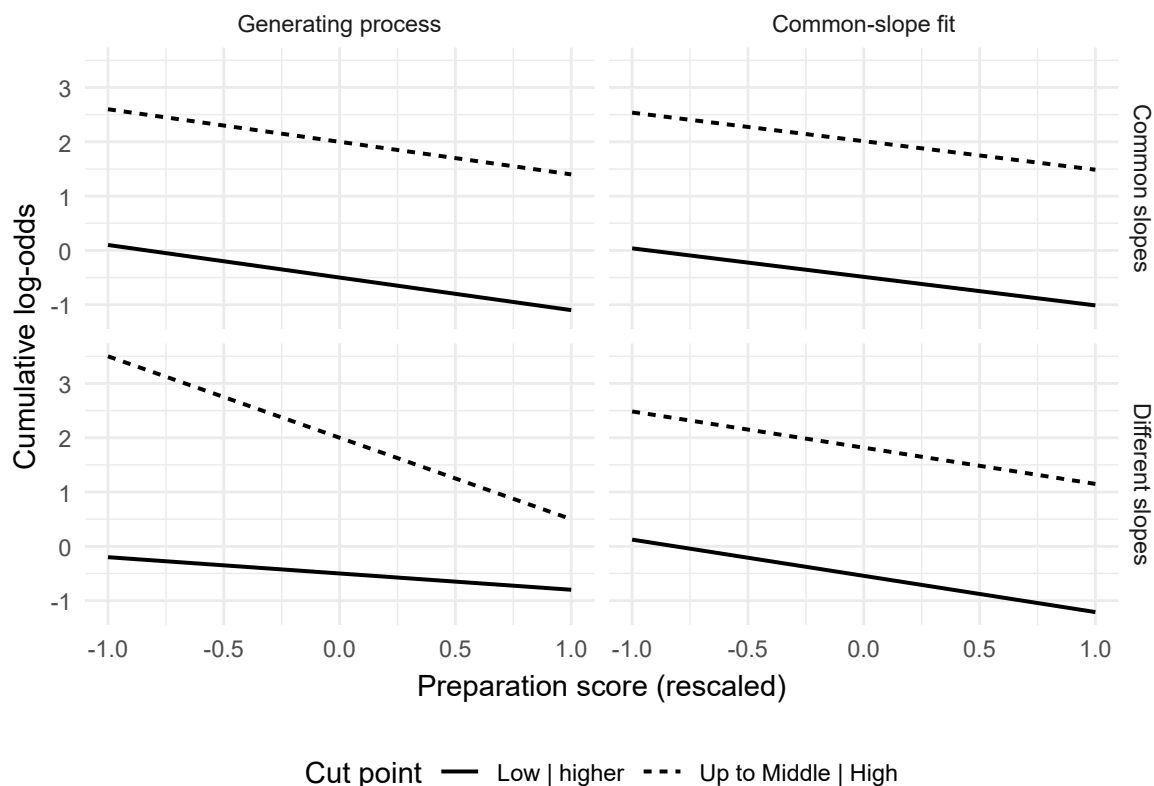


Figure 4: Known cumulative logits and the common-slope fits to two generated samples. The lower row makes the imposed restriction visible.

The fitted common slopes are .5253 and .6673. Sampling variation explains the first estimate's difference from its truth of .6. In the second sample, the model forces a compromise: its coefficient is a restricted likelihood summary, not an arithmetic average of the two true slopes. At $x = 1$, the second process has High probability .3775, whereas its common-slope fit gives .2406. A plausible-looking coefficient can thus conceal a consequential error on the response scale.

Freeing the slope at the second cut adds one parameter. The nested likelihood comparisons give $\chi^2(1) = .3466$, $p = .556$ for the first sample and $\chi^2(1) = 176.0519$, $p < .0001$ for the second.

The first result is compatible with common slopes; a nonsignificant test does not establish that restriction in an unknown population. For the second sample, the strong evidence and the probability discrepancy point in the same direction.

```
fit2 <- ordinal::clm(y ~ x, data = sim2)
free2 <- ordinal::clm(y ~ 1, nominal = ~x, data = sim2)
anova(fit2, free2)
```

Detection also depends on study size. A less obvious departure, $b_1 = .6$ and $b_2 = .85$, illustrates why a diagnostic p -value needs context. Hold that process fixed and generate 500 independent studies at each of two sample sizes. Repeat the same comparison under the true common-slope process as a reference.

Table 4

Repeated tests at the .05 level, with thresholds and predictor distribution held fixed. All fitted comparisons were valid; the 500 repetitions describe sampling variation across studies.

Process	Records per study	Rejections in 500 studies	Rejection proportion	Monte Carlo SE
Common slopes	300	25	.050	.0097
Common slopes	6,000	23	.046	.0094
Different slopes	300	75	.150	.0160
Different slopes	6,000	470	.940	.0106

The first two rows estimate false-positive rates when common slopes are true. The last two estimate power against this particular alternative. The same modest departure is detected much more often in the larger studies. More observations improve detection; they do not make the underlying departure larger. Conversely, a small study can miss a restriction that matters for its intended probability comparison.

Each reported proportion also has simulation uncertainty. Its Monte Carlo standard error is $\sqrt{\hat{r}(1-\hat{r})/500}$, where $\hat{r}$ is the rejection proportion. This describes the precision of the simulation's rate estimate, not the uncertainty of a coefficient in one study. One simulated sample supplies one test result; repeated samples are needed to estimate a rejection rate. In an empirical study, we still need the fitted category probabilities and substantive context to judge what a detected departure means.

The World Values Survey records order opinion as Too Little, About Right, Too Much government effort. Observed country proportions offer an early warning: Sweden's cumulative

logit differs from Australia's by .3826 at the lower cut but 2.009 at the upper. The gaps are not parallel. These are marginal summaries, so an adjusted model must still be checked rather than treating the graph as its formal test.

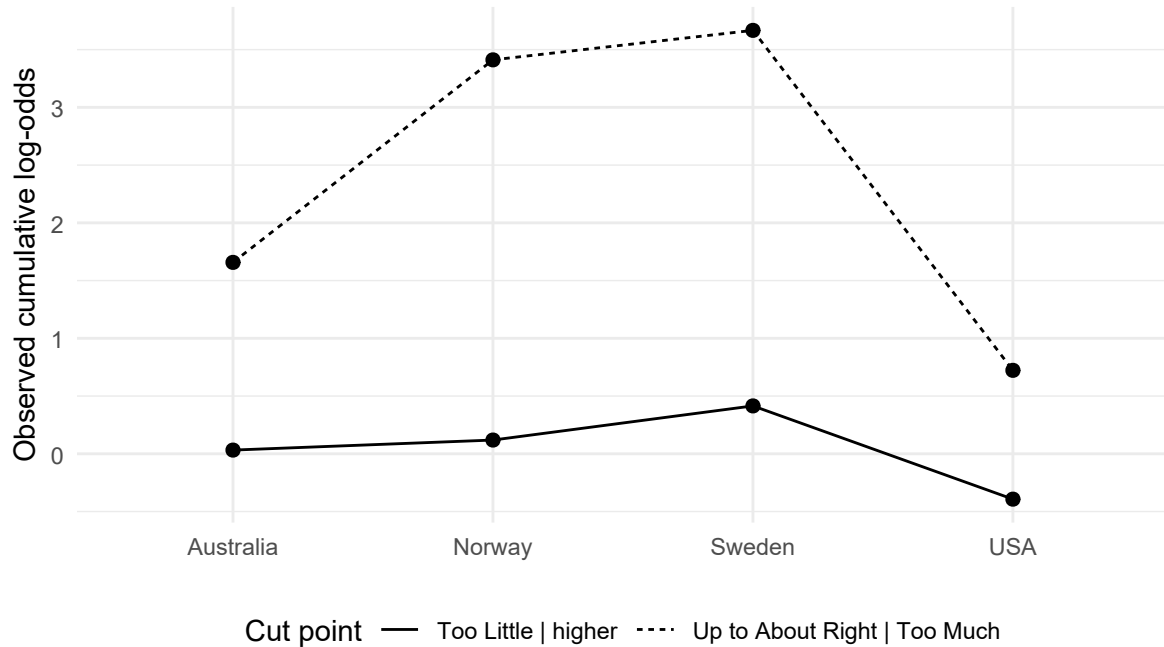


Figure 5: Observed cumulative logits by country, before adjustment.

3.3 Fit and interpret opinion among the recorded respondents

The four samples contain 1,874 Australian, 1,127 Norwegian, 1,003 Swedish and 1,377 United States respondents. The comparison adjusts for age, recorded gender, membership of a religion and university degree. There are few Too Much responses in Norway (36) and Sweden (25), a support feature that will matter when checking fitted composition.

```
po <- ordinal::clm(poverty ~ religion + degree + country + age + gender,
  data = opinion)
exp(confint(po, type = "Wald"))
```

Table 5

Proportional-odds fit for opinion about government effort; $-2\ell = 10402.5924$ on nine parameters, AIC 10420.5924. A positive coefficient shifts probability toward the higher ordered categories, from Too Little through About Right to Too Much.

Term	Coefficient	SE	OR	95% interval
Threshold Too Little About Right	0.7298	0.1041	—	—
Threshold About Right Too Much	2.5325	0.1103	—	—
Member of a religion	0.1797	0.0773	1.1969	[1.0285, 1.3928]
University degree	0.1409	0.0662	1.1513	[1.0112, 1.3108]
Norway against Australia	−0.3224	0.0738	0.7244	[0.6269, 0.8371]
Sweden against Australia	−0.6033	0.0795	0.5470	[0.4681, 0.6392]
United States against Australia	0.6178	0.0707	1.8548	[1.6149, 2.1303]
Age, per year	0.0111	0.0016	1.0112	[1.0081, 1.0143]
Male against female	0.1764	0.0530	1.1929	[1.0752, 1.3234]

Under this restriction, United States respondents have 1.8548 times the Australian respondents' cumulative odds above either cut at the same other predictor values: About Right/Too Much versus Too Little, and Too Much versus Too Little/About Right. It is not the odds of Too Much versus Too Little with About Right discarded.

Check the sign convention on a profile. For a 45-year-old religious woman without a degree in Norway, subtract the linear predictor from each threshold and apply the inverse logit. The cumulative probabilities are .5917 and .8979; subtracting gives category probabilities .5917, .3062 and .1021. A positive predictor shift decreases the cumulative probability below each cut and moves mass upward, as claimed.

```
profile <- data.frame(religion = "yes", degree = "no",
  country = "Norway", age = 45, gender = "female")
predict(po, profile, type = "prob")
```

3.4 Let the cut points have different contrasts

`nominal_test()` checks a predictor's proportional-odds restriction by allowing its effects to differ across cuts within the cumulative-link family. Country gives $\chi^2(3) = 362.3463$, $p < .0001$; degree gives 10.2866 ($p = .0013$), age 5.0587 ($p = .0245$), gender 3.0902 ($p = .0788$) and religion .414 ($p = .52$). These checks identify a large country departure and smaller possible departures; a large p-value does not establish a restriction.

```
ordinal::nominal_test(po)
cutfit <- ordinal::clm(poverty ~ religion + degree + age + gender,
  nominal = ~ country, data = opinion)
anova(po, cutfit)
```

Table 6

One common coefficient against two cut-specific ones, all in the direction of the higher ordered categories.

Country against Australia	Common log-odds		
	coefficient	Lower cut	Upper cut
Norway	−0.3224	−0.1227	−1.7812
Sweden	−0.6033	−0.4449	−2.0687
United States	0.6178	0.3626	0.8727

For Norway, the single common coefficient $-.3224$ masks a small lower-cut contrast and a much larger upper-cut contrast. The striking feature is how few Norwegian respondents choose Too Much. AIC falls from 10420.5924 to 10064.2461 for three more parameters; the fitted Too Much probability at the Norway profile falls from .1021 to .0289. This **partial proportional-odds** model frees country while retaining common slopes for the other predictors. Those remaining restrictions still require checking; freeing the largest departure does not settle every condition.

The paired bars hold age, religion, degree and gender fixed. Follow all three probabilities when the country restriction is relaxed: probability removed from Norway’s Too Much category must go somewhere else. In the United States panel the largest shift is in About Right. The figure shows how a restriction can alter the substantive description, not just a p -value. These are conditional profile comparisons, not country averages; better in-sample agreement does not establish the remaining restrictions.

A common-slope estimate after a demonstrated violation is at best a model-dependent compromise, not a literal arithmetic average of the cut-specific coefficients. Report cut-specific comparisons and fitted probabilities when they explain the substantive pattern.

The known-truth example in Section 3.2 separated detection from its consequences. Here the generating process is unknown, but the same interpretive step is available: inspect the probabilities at profiles that matter. The simulation’s rejection rates are not power estimates for these survey records. The Norwegian profile shows a practical consequence of this fitted restriction beyond the small p -value.

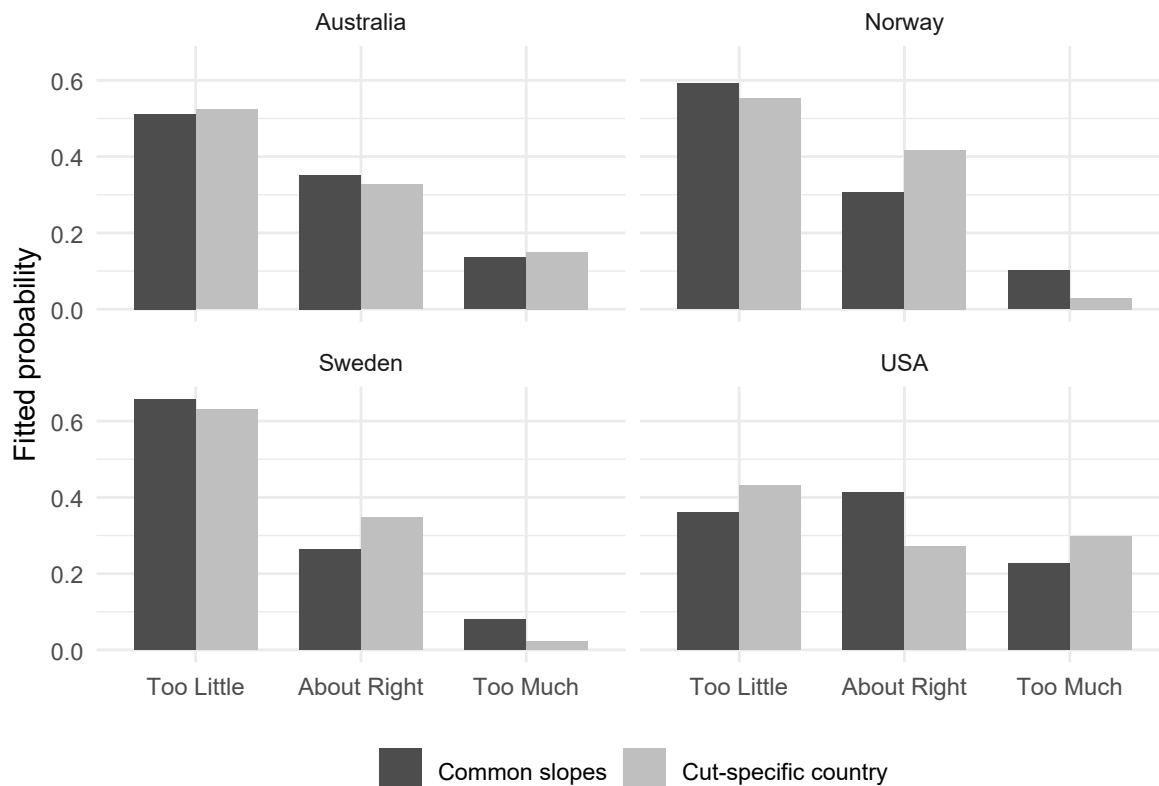


Figure 6: Relaxing the country restriction reallocates probability across the opinion categories.

3.5 Separate ordering from proportional odds

Return to employment and stipulate the order not working, part time, full time. The proportional-odds fit has four parameters, deviance 441.663 and AIC 449.663; the nominal fit has six parameters, deviance 422.8819 and AIC 434.8819. The lower nominal AIC favors its probability surface among these candidates on the same records.

These are generally **nonnested** families. Linear cumulative logits do not generally produce linear baseline-category logits, so subtracting the two deviances and calling the difference chi-square on two df is invalid. Valid common-slope checks compare cumulative models with common and cut-specific slopes: here income gives 4.1382 ($p = .0419$), and children 11.2982 ($p = .0008$). They challenge proportional odds, not the logical ordering of employment intensity.

```
dat$ord <- factor(dat$partic,
  levels = c("not.work", "parttime", "fulltime"), ordered = TRUE)
```

```
ord <- MASS::polr(ord ~ hincome + children, data = dat, Hess = TRUE)
AIC(ord, nom)
ordinal::nominal_test(ordinal::clm(ord ~ hincome + children, data = dat))
```

A nominal analysis remains natural for a question about distinct employment roles. If an ordered construct is substantively warranted, relaxing proportional odds can preserve that ordering. Conversely, good fit cannot make an arbitrary ordering meaningful. Collapsing adjacent categories also changes the measurement and estimand; a better statistic on the collapsed outcome cannot justify the merge because the outcome changed.

4 Counts and Rates: Poisson Regression

4.1 A positive mean with a specified variance

For count Y_i , the Poisson model assigns probabilities $P(Y_i = y \mid \mathbf{x}_i) = e^{-\mu_i} \mu_i^y / y!$ for integers $y \geq 0$. The **log link** connects its positive mean to a linear predictor:

$$\log \mu_i = \eta_i = \beta_0 + \sum_j \beta_j x_{ij}, \quad \mu_i = e^{\beta_0} \prod_j e^{\beta_j x_{ij}}. \quad (6)$$

Adding a coefficient on the log scale multiplies the expected count by its exponential. Unlike an unrestricted straight mean, this function cannot predict a negative count. A fitted mean can be fractional even though an observed count is an integer, because it is an expectation over repeated observations rather than one possible observation.

The Poisson distribution ties spread to the mean:

$$\text{Var}(Y_i \mid \mathbf{x}_i) = E(Y_i \mid \mathbf{x}_i) = \mu_i. \quad (7)$$

There is no separately fitted residual SD. As the mean rises the spread rises, while the distribution becomes less skewed. This variance condition will be the central problem in the absence data.

```
dpois(0:5, lambda = 2)
exp(c(0, 1, 2))
```

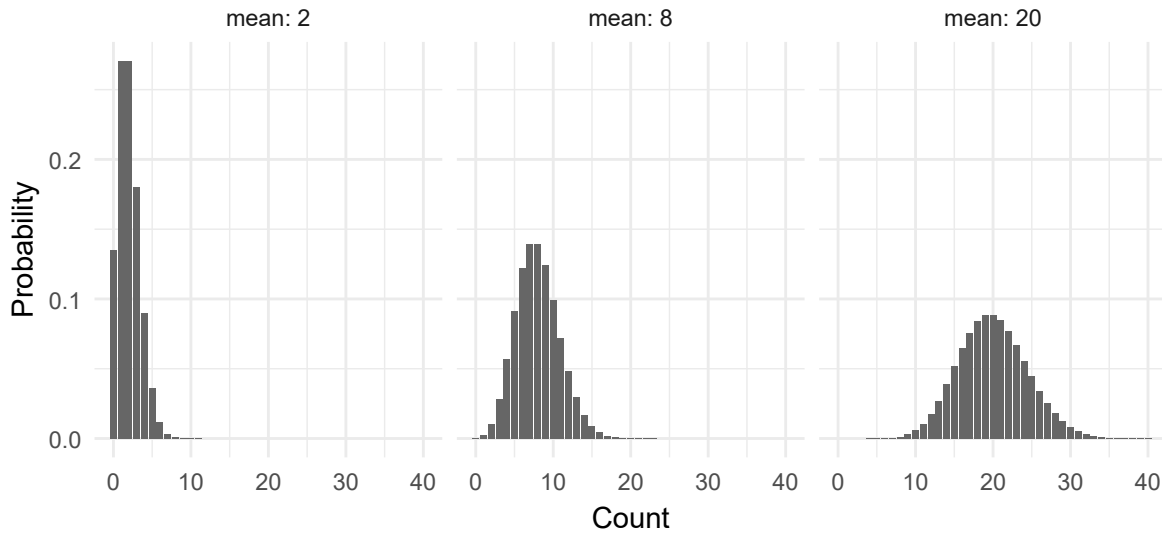


Figure 7: Poisson distributions at means 2, 8 and 20; the conditional variance equals the mean in each panel.

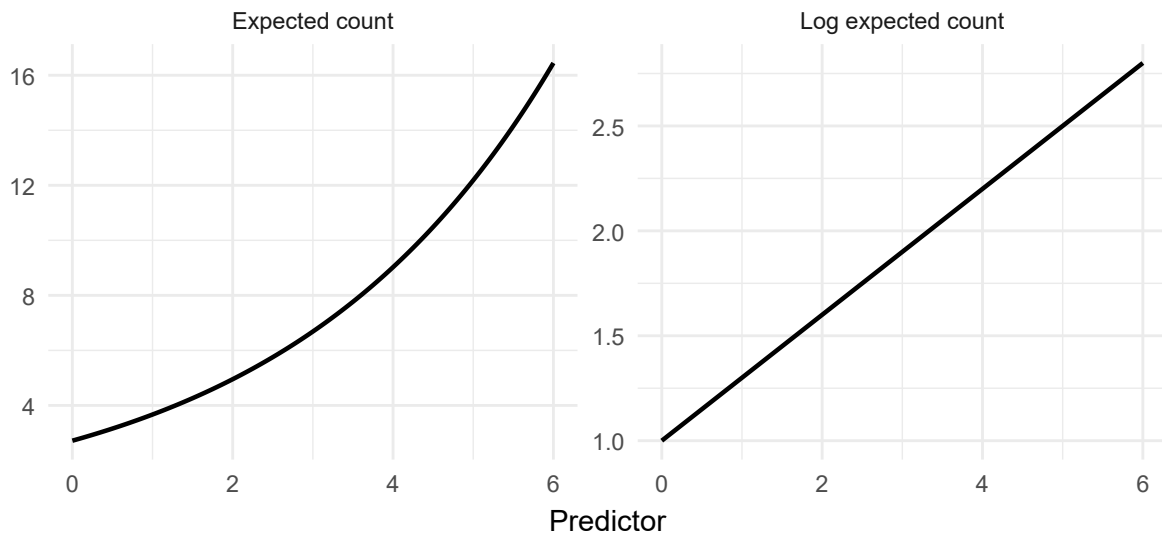


Figure 8: An additive log mean implies multiplicative expected counts.

4.2 Ratios and counts in the absence study

For one predictor unit at the same other predictor values, $\mu(x_j + 1)/\mu(x_j) = e^{\beta_j}$. This **incidence-rate ratio** (IRR) compares expected counts when windows are common, or expected

rates when exposure is accounted for. It is not a probability, an odds ratio or a within-person change. A constant ratio already implies larger absolute differences at larger baseline means; a ratio that itself changes with another predictor is an interaction on the log scale.

The **quine** study records days absent in one academic year. Its ethnic and learner classifications are the historical study’s own labels, not claims about inherent group qualities. We use the stated common academic year as the exposure, while recognizing that unrecorded enrollment or attendance-recording differences would undermine that comparison.

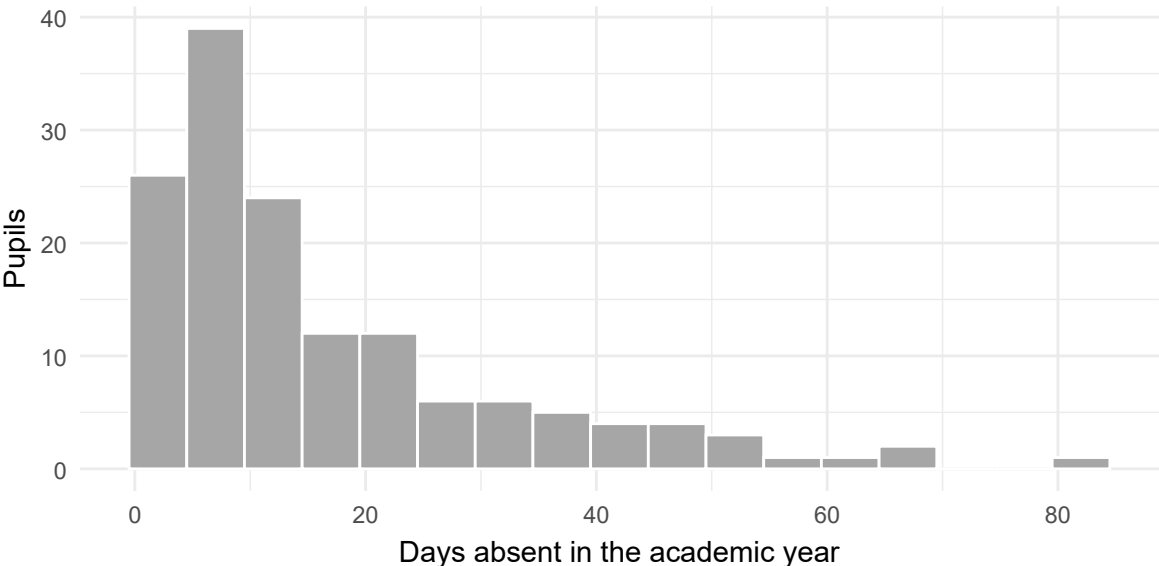


Figure 9: Recorded days absent among 146 pupils in a rural New South Wales study.

The observed mean is 16.4589 days, variance 264.1673, and quartiles 5, 11 and 22.75. That marginal variance is much larger than the marginal mean, but heterogeneous expected means alone can create such a difference. We must first condition on the predictors before diagnosing overdispersion.

Table 7
Variables and reference levels in the absence analysis.

Field	What it records	Reference level in the model
Days	Days absent in the academic year	Outcome count
Eth	Historical Aboriginal/not-Aboriginal classification	A: Aboriginal
Sex	Recorded sex	F: female
Age	School form, a four-level factor	F0: primary form
Lrn	Historical learner classification	AL: average learner

```
mp <- glm(Days ~ Eth + Sex + Age + Lrn,
  family = poisson(link = "log"), data = absence)
exp(coef(mp))
```

Table 8

Poisson fit for days absent; residual deviance 1696.7066 on 139 df, null deviance 2073.5328 on 145, AIC 2299.1836. Reference child: Aboriginal, female, primary form, average learner.

Term	Coefficient	SE	IRR
Intercept	2.7154	0.0647	15.1104
Not Aboriginal	−0.5336	0.0419	0.5865
Male	0.1616	0.0425	1.1754
Form 1	−0.3339	0.0701	0.7161
Form 2	0.2578	0.0624	1.2941
Form 3	0.4277	0.0677	1.5337
Slow learner	0.3489	0.0520	1.4176

The reference child’s fitted expected absence is 15.1104 days. The not-Aboriginal/Aboriginal IRR is .5865, at the same other recorded classifications. That ratio is interpretable only beside its baseline: it implies 8.862 expected days for a female average learner in F0 classified as not Aboriginal. These are conditional group descriptions, not effects of changing an identity or classification.

The output’s small p-values might initially appear impressive, but its residual deviance is 1696.7066 on 139 df. The conditional variance check in Section 5 shows why the usual Poisson standard errors are too small. The number of small p-values alone is not the diagnosis.

4.3 Exposure is part of the estimand

If unit i is observed for t_i days, person-years or another exposure, let λ_i denote its event rate. Then $\mu_i = t_i \lambda_i$. A log-linear rate model gives

$$\log \mu_i = \log t_i + \beta_0 + \sum_j \beta_j x_{ij}. \quad (8)$$

The exposure term has a coefficient fixed at one. Such a known-coefficient predictor is an **offset**. Doubling exposure doubles the expected count while retaining the same rate, a substantive restriction about how events accumulate. It is appropriate only when that relation matches the process and the exposure has been measured adequately.

Before fitting a rate model, name the event, the observation window and whether that window differs across units. Without the offset the coefficients compare counts over those potentially different windows; with it they compare rates. When exposure is related to predictors, omitting it can move slopes as well as the intercept. Even with equal exposure, naming it tells the reader what a count means.

An offset has no estimated coefficient, SE or p-value. To assess the proportional-exposure restriction, fit a free coefficient γ on $\log t_i$ and compare it with 1. Here γ is a locally defined exposure coefficient, distinct from the cumulative probabilities γ_{ij} in Section 3. The question is whether the stipulated accumulation rule is consistent with the data, not whether exposure is “significant” and can therefore be dropped.

4.4 A known process with unequal observation windows

The invented district example records behavioral referrals for 500 students. Higher screening risk makes mobility more probable, so students with higher rates tend to have fewer enrolled days. Mentoring is related to risk in the generated records. The declared conditional process is

$$\log \lambda_i = -4.2 - .35 \text{mentor}_i + .45 \text{risk}_i, \quad Y_i \sim \text{Poisson}(t_i \lambda_i). \quad (9)$$

The true mentoring and risk IRRs are .7047 and 1.5683. At seed 1202, 125 students have fewer than the full 180 days; exposure ranges from 31 to 180, correlating $-.2976$ with risk. The 79,090 enrolled days contain 1,195 referrals. Thus observation time and propensity for events must be separated to recover the declared rate comparison.

```
ratefit <- glm(events ~ mentor + risk, offset = log(days),
  family = poisson, data = sim)
countfit <- glm(events ~ mentor + risk, family = poisson, data = sim)
exp(coef(ratefit))
exp(coef(countfit))
```

Table 9

A declared process recovered with and without its exposure offset; 500 generated students, seed 1202.

Quantity	Truth	Offset model	Offset omitted
Mentoring IRR	0.7047	0.7024	0.7174
Risk IRR	1.5683	1.5700	1.4192
Intercept, daily rate	0.0150	0.0157	—
Residual deviance	—	521.9792	666.1568

Quantity	Truth	Offset model	Offset omitted
Pearson dispersion statistic	1	0.9438	1.2050

The offset fit recovers the declared IRRs: .7024 [.6202, .7955] for mentoring and 1.57 [1.4791, 1.6665] for risk. Omitting exposure moves the risk IRR to 1.4192, because less observed time partly conceals higher rates. The demonstration concerns a stipulated conditional process; it does not make observational mentoring coefficients causal in a real district.

A free exposure coefficient gives 1.1631 (SE .1124), with interval [.9429, 1.3833]. Testing $\gamma = 1$ gives Wald $z = 1.4517$, $p = .1466$, and a nested likelihood statistic 2.197, $p = .1383$. The records are compatible with proportional accumulation, while the interval describes the remaining uncertainty about that restriction.

```
freefit <- glm(events ~ mentor + risk + log(days),
  family = poisson, data = sim)
confint.default(freefit)["log(days)", ]
```

Table 10

Fitted rates at four profiles and the counts they imply over two observation windows; 500 generated students, seed 1202.

Profile	Rate per 100 enrolled days	Expected count, 180 days	Expected count, 60 days
Not mentored, average risk	1.5721	2.8298	0.9433
Mentored, average risk	1.1043	1.9877	0.6626
Not mentored, risk one SD higher	2.4682	4.4427	1.4809
Mentored, risk one SD higher	1.7337	3.1206	1.0402

The rates in a row are stable across windows, but the 180-day expected count is three times the 60-day count. One fitted number without its window would leave the study's main distinction invisible. These generated counts have marginal variance-to-mean ratio 1.599, yet the correctly specified conditional Pearson ratio is .9438. A marginal ratio above one therefore does not establish conditional overdispersion.

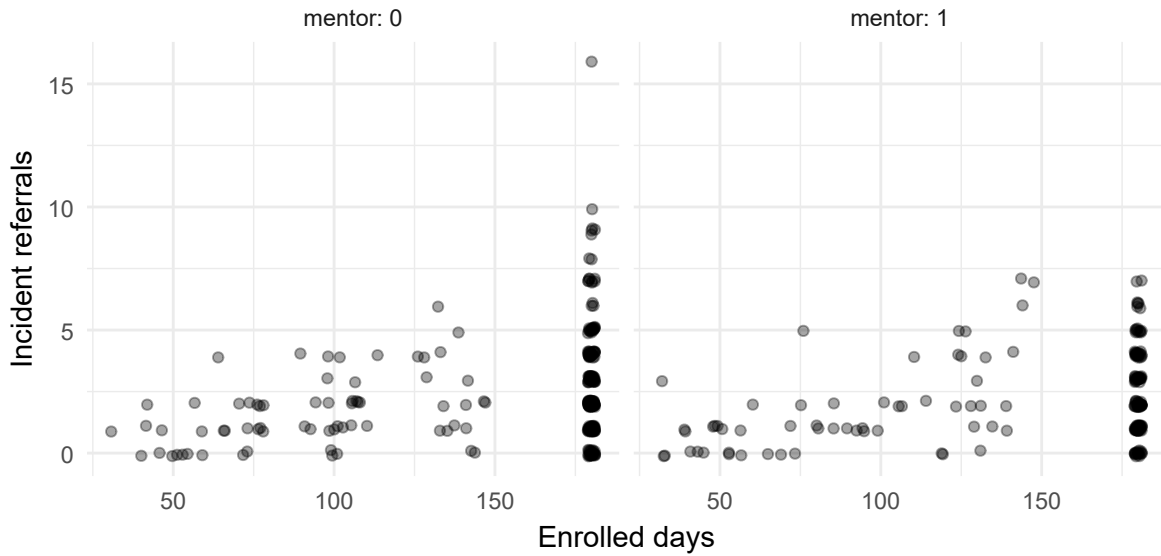


Figure 10: Counts depend on both rate and exposure in the invented 500-student study.

5 Overdispersion: Quasi-Poisson and the Negative Binomial

5.1 Compare spread around conditional means

Overdispersion means conditional variance exceeds the conditional mean, relative to the Poisson variance assumption. If k coefficients are estimated, a common diagnostic is the mean squared Pearson residual:

$$\hat{\phi} = \frac{1}{n - k} \sum_i \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}. \quad (10)$$

With an adequate Poisson mean and variance it estimates a value near one. The deviance ratio $D/(n - k)$ is a rough companion diagnostic, whose behavior also depends on the expected counts. It is not automatically a conservative or interchangeable estimate of dispersion.

```
sum(residuals(mp, type = "pearson")^2) / df.residual(mp)
deviance(mp) / df.residual(mp)
var(absence$Days) / mean(absence$Days)
```

The absence fit has Pearson ratio 13.1668, deviance ratio 12.2065 and marginal ratio 16.0501. The first two already condition on the four recorded predictors. Binning similar fitted means

makes the excess spread concrete, although bins still contain some mean heterogeneity and their raw variance-to-mean ratios are descriptive rather than exact tests.

Table 11

Observed mean and variance within five bins of Poisson-fitted values. Bins pool unequal fitted means, so their ratios are descriptive; some excess above 1 can come from remaining mean variation within a bin.

Fitted-value bin	Children	Observed mean	Observed variance	Variance / mean
1	36	8.0000	55.1429	6.8929
2	26	11.6538	91.8354	7.8803
3	33	19.6364	374.9886	19.0966
4	29	17.5172	193.8300	11.0651
5	22	29.8182	437.2987	14.6655

Before changing the variance family, consider the mean, exposure, recording process and dependence. All four predictors are factors, but the additive log mean still restricts interactions. Adding ethnicity by form improves deviance by 153.87 on three df and leaves a Pearson ratio of 11.7342. This particular interaction explains little of the excess spread; it does not prove every other mean specification correct. The comparison below retains the additive mean as a working approximation while examining what variance assumptions do to uncertainty.

If the mean is correct and independent-row regularity conditions hold, Poisson estimating equations can consistently estimate its coefficients even with a wrong variance (Cameron & Trivedi, 2013). The conventional model-based SEs can then be seriously wrong. Under variance $\phi\mu$, a variance multiplier near 13 makes those SEs too small by about $\sqrt{13}$. That mechanism, rather than impressive-looking p-values, motivates a remedy.

5.2 Quasi-Poisson preserves the fitted mean

Quasi-Poisson keeps the same log mean and specifies $\text{Var}(Y_i | \mathbf{x}_i) = \phi\mu_i$. Its coefficient estimating equations match the Poisson equations, so the estimates are identical; SEs multiply by $\sqrt{\hat{\phi}}$. Here the multiplier is 3.6286. R's summary coefficient tests use an approximate t reference. The reported `confint.default()` intervals instead use the large-sample Normal Wald reference, a distinction that should remain explicit.

```
mq <- glm(Days ~ Eth + Sex + Age + Lrn,
  family = quasipoisson, data = absence)
exp(confint.default(mq))
drop1(mq, test = "F")
```

The not-Aboriginal IRR stays .5865, while its interval widens from [.5403, .6367] to [.4354, .7900]. None of the three individual form contrasts excludes one, yet the whole form block gives $F(3, 139) = 4.5966$, $p = .0042$. A joint factor comparison and a single contrast against one arbitrary reference answer different questions.

Quasi-likelihood specifies no complete outcome distribution. Consequently there is no ordinary log-likelihood, AIC or BIC, and R returns NA for AIC deliberately. Nested quasi-Poisson mean models can be compared using approximate scaled-deviance F tests. A predictive count distribution also needs assumptions beyond this mean-and-variance specification.

5.3 Negative binomial changes the distribution and weighting

The negative binomial supplies a full distribution with

$$E(Y_i | \mathbf{x}_i) = \mu_i, \quad \text{Var}(Y_i | \mathbf{x}_i) = \mu_i + \frac{\mu_i^2}{\theta}, \quad \theta > 0. \quad (11)$$

One interpretation mixes Poisson rates across units sharing the same observed predictors, with a gamma mixing factor of mean one. That gives quadratic extra variance in the mean (Hilbe, 2011). It is a possible mechanism, not something uniquely established by a fitted negative-binomial curve. Small θ means more overdispersion; as θ grows, the variance approaches Poisson.

Quasi-Poisson has constant variance-to-mean multiplier ϕ , whereas the negative binomial's multiplier $1 + \mu/\theta$ increases with the mean. Thus the two families disagree about how variability grows across profiles, not merely about one correction factor.

```
mn <- MASS::glm.nb(Days ~ Eth + Sex + Age + Lrn, data = absence)
mn$theta
exp(confint.default(mn))
```

The fit gives $\hat{\theta} = 1.2749$ (SE .161) and Pearson ratio .9912. Large fitted means receive less information weight relative to Poisson, so coefficients change as well as SEs. The not-Aboriginal IRR becomes .5659 with interval [.4190, .7643]. This is a new likelihood fit, not the old fit with a rescaled covariance matrix.

5.4 Compare remedies against the question and the evidence

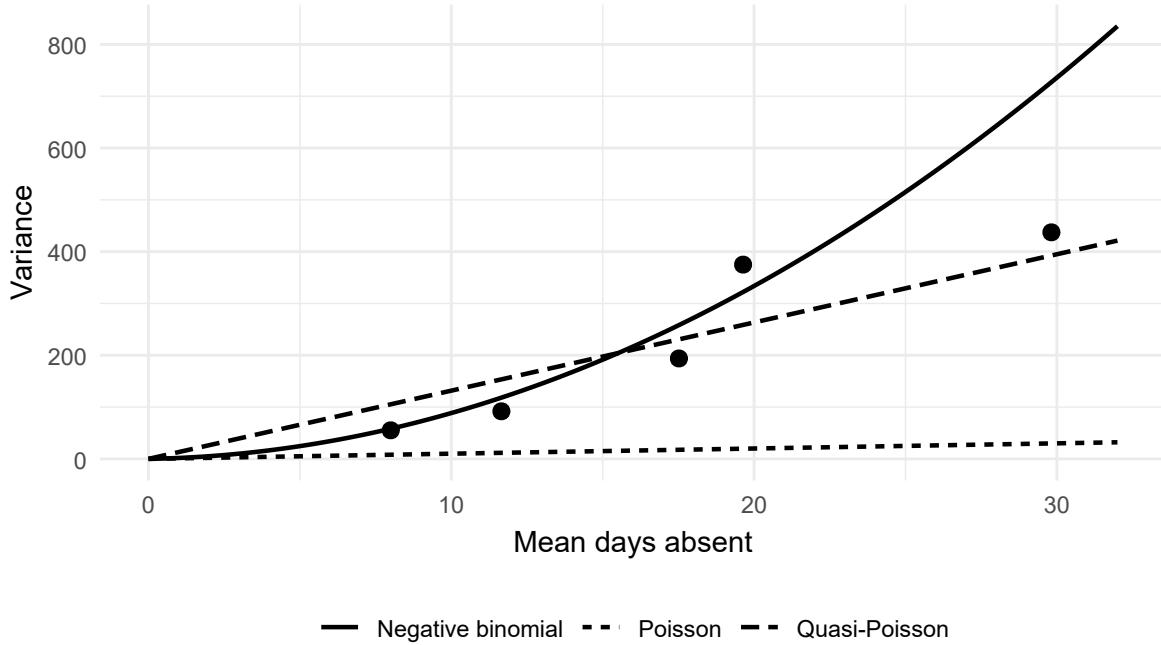


Figure 11: Variance functions fitted to the absence records, with descriptive binned means and variances.

Table 12

Days absent under three variance specifications on the same 146 records. SEs are on the log-coefficient scale, not the IRR scale. The conditional-dispersion evidence motivates comparing uncertainty specifications.

Term	IRR, Poisson and quasi	SE, Poisson	SE, quasi	IRR, negative binomial	SE, negative binomial
Intercept	15.1104	0.0647	0.2347	18.0759	0.2284
Not Aboriginal	0.5865	0.0419	0.1520	0.5659	0.1533
Male	1.1754	0.0425	0.1543	1.0858	0.1599
Form 1	0.7161	0.0701	0.2543	0.6386	0.2397
Form 2	1.2941	0.0624	0.2265	1.0921	0.2362
Form 3	1.5337	0.0677	0.2456	1.4289	0.2483
Slow learner	1.4176	0.0520	0.1888	1.3392	0.1865

The SE columns are on the coefficient's log scale, not the IRR scale. Quasi-Poisson multiplies them all by 3.6286; negative binomial changes them by different factors. Under both, ethnicity is the only individual contrast whose 95% interval excludes one. Form remains jointly informative: negative-binomial deletion tests give approximately $\chi^2(3) = 11.979$, $p = .0075$. These package

deletion tests hold its fitted dispersion parameter fixed; refitting dispersion in every reduced model is a different likelihood comparison and should be labeled as such.

Table 13

Choosing a variance specification from the target, the mechanism and the checking evidence.

Response	Target it serves	Plausible mechanism	Checking evidence
Quasi-Poisson	The same mean model with honest uncertainty, where no likelihood-based comparison is needed	A constant variance-to-mean multiplier across the fitted range	$\hat{\phi}$ far above 1, with the binned variance-to-mean ratio roughly level in the fitted mean
Negative binomial	A full likelihood, for information criteria, intervals or a predicted count distribution	The Poisson mean varies across units sharing predictor values, so variability grows with the mean	$\hat{\phi}$ far above 1, with the binned ratio rising in the fitted mean, and a Pearson statistic near 1 after the refit
Revise the mean instead	The same question, answered with a mean the records support	An omitted predictor, an unmodelled interaction, or a missing exposure	Enriching the mean removes most of the excess spread

AIC is 2299.1836 for Poisson and 1109.151 for negative binomial; there is no quasi-Poisson entry. The Poisson limit is $1/\theta = 0$, on the boundary of the negative-binomial parameter space. Under appropriate regularity conditions the likelihood-ratio null reference is a half-mass at zero and half χ_1^2 mixture, not ordinary chi-square on one df. Here the large likelihood improvement accompanies compelling dispersion evidence; it does not excuse checking mean structure or dependence. Other mixture or zero-inflation comparisons can have further unidentified parameters, so this exact mixture must not be assumed universal.

Before reading the diagnostic numbers, predict which model can place more probability in both tails while retaining similar mean comparisons.

A fitted count mean does not show how much probability the model assigns to zero or to a long upper tail. For each pupil, the fitted model assigns a probability to each count band. Adding those probabilities across pupils gives the expected number in that band. This preserves the pupils' different fitted means. Substituting one overall mean would instead create a different distribution.

```
sum(dpois(0, fitted(mp)))
sum(dnbinom(0, mu = fitted(mn), size = mn$theta))
sum(ppois(50, fitted(mp), lower.tail = FALSE))
```

Table 14

Observed and fitted count-band totals; the final band includes the entire upper tail.

Band	Observed	Poisson	Negative binomial
0	9	0.0141	6.2136
1-2	10	0.4492	14.4933
3-5	26	5.2770	21.0057
6-10	24	29.6620	28.4776
11-20	35	70.6830	35.3951
21-50	34	39.9072	33.0525
51 or more	8	0.0074	7.3623

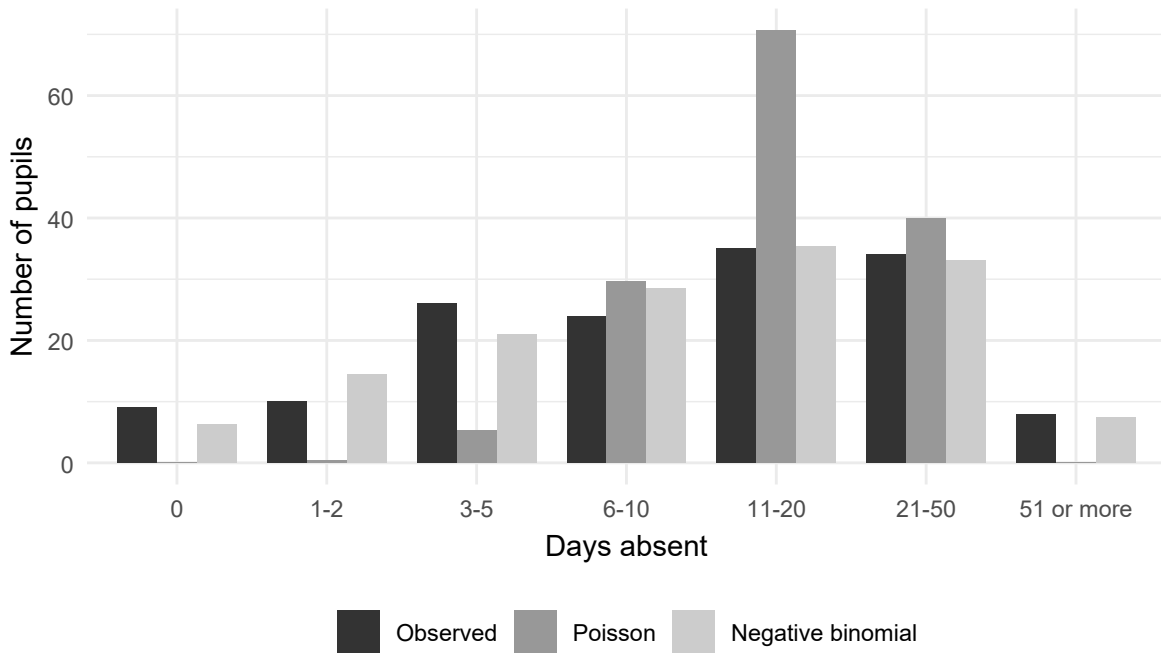


Figure 12: Observed and fitted count-band totals show where the two count distributions differ.

The Poisson fit expects almost no zero counts and almost no counts above 50; nine and eight were observed. The negative-binomial distribution places appreciable probability in both regions. Its total expectations sum to 146, just as the observed counts do. Summing only counts from 0 to 100 would silently lose some of its upper tail, so the last band is calculated with the distribution function all the way to infinity.

These are fitted, model-dependent comparisons. Their discrepancy is a useful diagnostic, not automatically a chi-square test with a degrees-of- freedom count obtained by subtracting

arbitrary numbers of bins and parameters. Several expected Poisson totals are very small, and every pupil has a different fitted distribution. A reference distribution for a formal statistic would need an argument or a simulation reflecting this fitting procedure.

Quasi-Poisson specifies a mean and a variance, not a full count distribution. There is therefore no corresponding quasi-Poisson column of expected band probabilities without adding a distributional assumption. This is the same reason its fitted variance alone does not supply a likelihood-based AIC.

6 Check and Compare Across the Family

6.1 Residuals use a model-specific yardstick

For counts, a Pearson residual is $(y_i - \hat{\mu}_i) / \sqrt{V(\hat{\mu}_i)}$; a signed deviance residual uses the square root of that observation's deviance contribution. They are read against their own response family, not against an assumed Normal residual cloud. The largest absolute Pearson residual in the absence fit falls from 13.5933 under Poisson to 3.7975 under negative binomial; the corresponding maximum deviance residual falls from 9.9091 to 2.7918. The pupil absent 81 days has fitted means 27.7199 and 26.4371: the large change is the variance yardstick, not a disappearance of the unusual observation.

6.2 Compare observed and fitted composition

For categories, group records meaningfully and sum fitted probabilities to obtain expected counts. The descriptive discrepancy $X^2 = \sum (O - E)^2 / E$ shows where a model misses. Automatically attaching a chi-square reference is risky when continuous predictor values are pooled into coarse cells, parameters were estimated on ungrouped data, or expected cells are small.

For the nominal labour-force fit, total fitted category counts match 155, 66 and 42 to numerical precision because of the intercept score equations. Within eight income-band-by-children groups, $X^2 = 12.4195$, with smallest fitted cell 1.0738. Read that as a display diagnostic, not a calibrated goodness-of-fit test with a mechanically subtracted df. The empty high-income full-time cell with children present remains a support question even if overall composition looks reasonable.

Table 15

Observed and fitted counts in the highest ordered category, by country. Over all twelve country-by-category cells, $X^2 = 322.9447$ under proportional odds and 0.0164 under the cut-specific model.

Country	Observed Too Much	Proportional odds	Cut-specific
Australia	300	271.1860	300.0223
Norway	36	126.8882	36.3260
Sweden	25	89.7115	24.9734
United States	450	344.1366	448.1760

The proportional-odds fit expects about 127 Norwegian and 90 Swedish Too Much responses where 36 and 25 occurred. The cut-specific model brings those expectations much closer. This discrepancy shows what the restriction was concealing on the scale of the data. Close in-sample agreement after adding flexibility is useful but is not held-out validation.

```
vars <- c("religion", "degree", "country", "age", "gender")
P <- predict(po, newdata = opinion[, vars], type = "prob")$fit
colSums(P)
table(opinion$country, opinion$poverty)
```

6.3 Match comparison rules to the fitted models

AIC and BIC use $-2\ell + 2k$ and $-2\ell + k \log n$, counting all estimated parameters. Nominal slopes enter $K - 1$ equations; cumulative models count thresholds; negative-binomial criteria count dispersion as well as mean coefficients. Compare likelihoods only for the same records, outcome coding and likelihood convention.

Table 16

Likelihood criteria within each set of records. Check nesting separately; count-model deviances use different saturated likelihoods.

Records and outcome	Model	k	Deviance or -2ℓ	AIC
263 women, three participation categories	Null	2	500.4926	504.4926
	Husband's income only	4	486.4405	494.4405
	Children only	4	438.0351	446.0351
	Proportional odds, both predictors	4	441.6630	449.6630

Records and outcome	Model	k	Deviance or -2ℓ	AIC
5,381 respondents, three ordered categories	Generalized logits, both predictors	6	422.8819	434.8819
	Null	2	10740.3765	10744.3765
	Without country	6	10653.4731	10665.4731
146 pupils, days absent	Proportional odds	9	10402.5924	10420.5924
	Cut-specific country	12	10040.2461	10064.2461
	Poisson	7	1696.7066	2299.1836
	Quasi-Poisson	7	1696.7066	none
	Negative binomial	8	167.9518	1109.1510

Within the women block, the proportional-odds and generalized-logit rows can be compared by AIC but are not a nested chi-square comparison. Within the opinion block, freeing country slopes is a nested comparison in the same cumulative-link family. In the count block, Poisson and negative-binomial deviances use different saturated likelihoods: their deviance difference is not their likelihood-ratio statistic. Their AICs use comparable complete likelihoods, with the boundary qualification for a formal test. Nothing is comparable across the three datasets' blocks.

6.4 What changing a family cannot repair

A variance remedy cannot supply a missing exposure, recover an unmeasured predictor, undo measurement error or establish independent sampling. Pupils within schools, patients within clinics and repeated observations need dependence-aware uncertainty or models. Counts truncated or censored by a recording rule need that rule represented; an estimated population denominator adds uncertainty that a fixed offset does not include. The choice set in a nominal model and the substantive order in an ordinal model are also claims about measurement, not facts certified by a small residual or a favorable criterion.

7 From Analysis to a Defensible Claim

7.1 Use the outcome scale alongside the ratio

A report states the analytic units and question, response model and coding, ratio and its interval, fitted probabilities or counts at named profiles, condition checks, and limits of extension. These are pieces of a reasoned account, not a required six-sentence ritual. A fitted probability without an interval is a point summary, not an exact quantity; an interval on a coefficient elsewhere in the paragraph does not quantify every derived probability or count. Evaluating a rule on new cases is a separate task from illustrating its fitted means.

7.2 Three reports, each bounded by its study

Labour-force composition

Among the 263 women recorded in the 1977 Canadian extract, an additional thousand dollars of husband's income was associated with a full-time-to-not-working RRR of .9073 (95% interval [.8587, .9587]), adjusting for children present. With no children present, fitted full-time probabilities were .7817 at income 5 and .3323 at 25 thousand dollars. Part-time comparisons were less precisely determined. A nominal composition model had lower AIC than the candidate proportional-odds model, whose common-slope restriction was also challenged by within-family tests. These conditional comparisons do not identify consequences of changing household income or childbearing.

Opinion about government effort

Opinion was ordered from Too Little through About Right to Too Much in 5,381 recorded respondents. A common-slope cumulative model gave a United States/Australia odds ratio of 1.8548 [1.6149, 2.1303] above either cut, but country effects differed substantially across cuts ($\chi^2(3) = 362.3463$, $p < .0001$). Allowing cut-specific country coefficients reduced AIC from 10420.5924 to 10064.2461 and better represented the few Too Much responses in Norway and Sweden. These are conditional cross-sectional comparisons. The extract does not supply survey-design information, and remaining common-slope restrictions require further checking.

Absence counts

Days absent during a common academic year were modeled for 146 recorded pupils. The Poisson Pearson dispersion ratio was 13.1668; one additional interaction left 11.7342. A negative-binomial fit gave $\hat{\theta} = 1.2749$ (SE .161) and Pearson ratio .9912. The not-

Aboriginal/Aboriginal IRR was .5659 [.4190, .7643] at common recorded sex, form and learner classification. For female average learners in F0, fitted expectations were 18.0759 and 10.2288 days respectively. A quasi-Poisson fit gave a similar substantive reading with wider uncertainty than Poisson. The historical group labels describe this study; the comparison does not identify an effect of changing a classification, and dependence or unrecorded differences remain possible.

7.3 Diagnose an overclaim before calculating again

“The relative-risk ratio was .9073” omits the compared categories. “Swedes had lower odds” omits the cumulative direction. “The offset was not significant” mistakes a fixed coefficient for an estimate. “The model predicted well” confuses in-sample fit with evaluated performance. Each repair restores the missing statistical meaning, not merely a cautious adjective.

An interval covering one is also a quantitative result. The negative-binomial male/female IRR interval [.7936, 1.4855] permits substantively lower or higher expected absence. Whether that range is small enough for a decision depends on the question, not only on its p-value (Wasserstein & Lazar, 2016).

8 Reading Bridge

Nominal models explain a composition through several baseline-category logits; ordinal models use cumulative probabilities and testable slope restrictions; count models connect positive means to exposure and a variance assumption. Their ratios are different quantities, each needing an outcome-scale comparison and a stated scope. Chapter 13 changes the goal: evaluating how a fitted rule performs on records it has not seen.

Key terms

These are the concepts introduced here. The Part One Concept Reference locates likelihood, links, model comparison and other earlier concepts.

Table 17

Concepts introduced here.

Term and common alternatives	Meaning or essential distinction
Outcome triage	The measurement judgment, made before any model is fitted, classifying an outcome as binary, nominal, ordered, a count, a rate, a proportion of known trials, or a duration. The model follows the classification and cannot establish it.
Generalized logit model (baseline-category logit; multinomial logistic regression)	For K unordered categories, $K - 1$ log-ratio equations against one baseline, each with its own intercept and slopes. The fitted probability surface is the same whichever baseline is chosen; the coefficients are not.
Baseline outcome category	The outcome category in the denominator of every generalized-logit equation: a coordinate choice, distinct from the reference level of a categorical <i>predictor</i> , whose change alters every reported ratio.
Relative-risk ratio (RRR)	The exponentiated generalized-logit coefficient: the factor by which one category's probability relative to the baseline's differs per predictor unit. It is not a probability difference, and is incomplete unless both compared categories are named.
Independence of irrelevant alternatives (IIA)	The generalized logit's assumption that the ratio of two categories' probabilities does not depend on which other categories were available: a claim about the choice set, argued rather than settled by one test.
Cumulative probability and cumulative logit	$\gamma_{ij} = P(Y_i \leq j)$ at each of the $K - 1$ cut points, and its log-odds. A category probability is a difference of two adjacent cumulative probabilities.
Threshold (cut point; ordinal intercept)	One of the ordered constants $\zeta_1 < \dots < \zeta_{K-1}$ placing the cut points on the logit scale. Their ordering keeps fitted probabilities nonnegative; they are not comparisons and are not read as effects.
Proportional odds (common-slope restriction; partial proportional odds)	The restriction that one slope vector governs every cut point, so a single odds ratio applies at all of them. It is imposed by the analyst, not conferred by ordering the levels, and is testable against a cut-specific alternative.

Term and common alternatives	Meaning or essential distinction
Poisson regression (log-linear count model)	A GLM with a Poisson random component and a log link, so the conditional mean is positive and differences are multiplicative on the count scale. Its conditional variance is not free: it is forced to equal the mean.
Incidence-rate ratio (IRR; rate ratio)	The exponentiated coefficient of a count or rate model: the factor by which the expected count, or rate when an exposure is included, differs per predictor unit. It hides its baseline, so an absolute difference is not recoverable from it alone.
Exposure and offset	The window over which each unit was observed, and $\log t_i$ entered with its coefficient fixed at one so the model describes a rate. An offset is a decision about the estimand, not an estimate: no SE, no p -value, and freeing its coefficient gives a testable alternative.
Overdispersion and the dispersion statistic	Conditional variance exceeding the conditional mean, assessed by the mean squared Pearson residual $\hat{\phi}$ under a suitable mean model; the deviance ratio is a rough diagnostic and need not be near 1 for sparse counts even under a correct Poisson model. The diagnosis is conditional; a marginal ratio above one is not evidence.
Quasi-Poisson (quasi-likelihood; scaled Poisson)	The Poisson mean with variance $\phi\mu$: identical estimates and every SE multiplied by $\sqrt{\hat{\phi}}$. No full distribution is specified, so there is no log-likelihood and no AIC or BIC.
Negative-binomial regression	A count model with variance $\mu + \mu^2/\theta$, arising when the Poisson mean varies across units sharing predictor values. A different likelihood, not a rescaling: estimates move because large fitted means are down-weighted, and the Poisson comparison sits on the boundary $1/\theta = 0$.

Further reading

Agresti (Agresti, 2010, 2018) develops categorical models; McCullagh (McCullagh, 1980) gives the cumulative-link framework. Cameron and Trivedi (Cameron & Trivedi, 2013) and Hilbe (Hilbe, 2011) develop counts and dispersion. The `ordinal`, `nnet` and `MASS` documentation gives the parameterizations used here (Christensen, 2025; Venables & Ripley, 2002).

Exercises

Exercise 1: Triage four outcomes and find the exposure (Foundational).

For each of the following, state the outcome kind from the chapter’s triage table, name the model family, and say in one sentence what the exposure is or why none is needed. (1) The field of study a first-year student enrolls in, from six faculties. (2) A patient’s self-rated recovery, recorded as worse, unchanged, or better. (3) The number of behaviour referrals recorded for each of 400 pupils during one full school year. (4) The number of emergency presentations from each of 60 districts during 2023, where the districts have populations from 4,000 to 210,000. (5) A pupil’s score out of 20 on a spelling test. For (4), write the linear predictor with its offset term and say which quantity the coefficients then compare.

Exercise 2: Read a generalized-logit result (Foundational).

A generalized-logit model of labour-force participation among 263 women, with not working outside the home as the baseline, reports relative-risk ratios of 0.9073 per thousand dollars of husband’s income and 0.0774 for children present in the full-time equation, and 1.0069 and 1.0217 in the part-time equation, with intervals [0.8587, 0.9587], [0.0381, 0.1574], [0.9617, 1.0543] and [0.4075, 2.5619]. (1) Write one complete sentence reporting the 0.9073, naming both compared categories. (2) State what the part-time equation does and does not establish. (3) A colleague reports “the relative-risk ratio for children was 0.0774” with no other wording. Say what is missing and why the number cannot be checked without it. (4) The same data refitted with full-time work as the baseline give a children coefficient of 2.5586 in the not-working equation. Explain why the two fits are the same model.

- (5) Use the chapter’s fitted-probability curves to explain why a constant full-time versus not-working relative-risk ratio does not give a constant full-time probability difference. Predict what happens to all three curves when the reference category changes.

Exercise 3: State the cumulative direction (Foundational).

A cumulative-logit model of opinion about government effort, coded so that a positive coefficient shifts probability toward the higher ordered categories from Too Little through About Right to Too Much, reports thresholds 0.7298 and 2.5325 and coefficients 0.1797 (religion), -0.3224 (Norway against Australia), 0.6178 (United States against Australia) and 0.0111 per year of age. (1) Report the United States odds ratio of 1.8548 in one sentence with its direction and its conditioning set. (2) Say what the Norway coefficient means in the same terms. (3) Calculate the fitted probability of answering Too Little for a 45-year-old religious woman without a degree in Norway, using $\text{logit}(\gamma_1) = \zeta_1 - \mathbf{x}^\top \boldsymbol{\beta}$ and the coefficients above. (4) Explain in one sentence why the thresholds are not reported as effects.

- (5) The common-slope model gives Too Much probability .1021 at this Norway profile, whereas the partial model gives .0289. Use the paired probability display to explain what the common restriction was concealing, and name one reason why the better fitted picture does not settle the analysis. Predict what would happen to the frequency of rejection if a fixed modest slope departure were studied with 6,000 rather than 300 records. Explain why this change would not by itself show a larger practical consequence.

Exercise 4: The offset is a decision, not an estimate (Foundational).

Five hundred students were observed for between 31 and 180 enrolled days, and behavioural incidents were counted. Two Poisson models were fitted to the same records: one with $\log(\text{days enrolled})$ as an offset, giving a risk incidence-rate ratio of 1.57, and one without it, giving 1.4192. (1) State the estimand of each model in one sentence each. (2) The data were generated with a true risk rate ratio of 1.5683. Explain why omitting the offset moved a slope and not only the intercept. (3) A reviewer asks for the offset's standard error and p -value. Explain why there are none, and describe the model that would answer the underlying question. (4) At a fitted rate of 1.5721 incidents per 100 enrolled days, give the expected number of incidents over 180 days and over 60 days, and say which of the three numbers is the stable quantity.

Exercise 5: Diagnose dispersion before choosing a family (Integrative).

A Poisson regression of days absent on four categorical predictors, fitted to 146 children, has residual deviance 1696.7066 on 139 degrees of freedom and a Pearson dispersion statistic of 13.1668; every coefficient has $p < .001$. (1) State what the dispersion statistic estimates under a correct Poisson model, and by what rescaling factor the quasi-Poisson variance model would imply. (2) Adding an ethnicity-by-form interaction improves the deviance by 153.87 on three degrees of freedom but leaves the dispersion statistic at 11.7342. Explain what this says about that particular interaction and what it leaves unresolved. (3) A quasi-Poisson refit leaves the not-Aboriginal incidence-rate ratio at 0.5865 and raises its log-coefficient standard error

from 0.0419 to 0.1520; a negative-binomial refit gives 0.5659 with log-coefficient standard error 0.1533. Explain why one estimate moved and the other did not. (4) The quasi-Poisson fit has no AIC and the negative-binomial fit has AIC 1109.151 against the Poisson 2299.1836. Say which comparisons are legitimate and which is not, and why.

- (5) Nine pupils have zero absences and eight exceed 50 days. Poisson expects .0141 and .0074 in these bands; negative binomial expects 6.2136 and 7.3623. Explain what this adds to the coefficient and AIC comparisons, and why there is no comparable quasi-Poisson probability column without a further assumption.

Exercise 6: Critique a categorical-outcome report (Integrative).

A report states: “Multinomial logistic regression showed that higher household income reduced the relative-risk ratio to 0.91. An ordinal model showed that Swedish respondents had significantly lower odds ($OR = 0.55$). A Poisson model of clinic visits showed that every predictor was highly significant ($p < .001$), and the exposure offset was retained because it was significant.” (1) Name five distinct omissions or unsupported readings, using the chapter’s vocabulary. (2) Rewrite the passage as three defensible sentences, inventing no new numbers but adding the information each claim requires. (3) State one check the authors should have reported for each of the three models, and say what each check would and would not establish.

Appendix A: Choice Sets and Cut-specific Structure

What a nominal choice model assumes

In a baseline-category multinomial logit, the probability ratio for two categories depends on their two linear predictors. Adding a third category changes the denominator of both probabilities, so it leaves their ratio unchanged if those predictors stay fixed. This property is called **independence of irrelevant alternatives (IIA)**.

Imagine a travel choice between a car and a blue bus. If an otherwise identical red bus becomes available, people who preferred the bus might split between the two colours, leaving car use nearly unchanged. A model that treats red and blue buses as independent alternatives instead redistributes probability according to a more restrictive rule. The problem is substantive similarity between alternatives, rather than a minor numerical imperfection. The same question matters if two response categories are nearly interchangeable or the available choice set varies between people.

For labour-force participation, consider whether part-time and full-time work are substitutes in the same way as not working and either work category. The observed labels alone do not answer that question. IIA checks and more flexible choice models can be useful, but a small-sample

formal test cannot establish a plausible choice mechanism. The nominal fit in this chapter is a conditional description of the recorded composition, with the stated working structure.

A partial proportional-odds comparison

The opinion example can be fitted through `MASS::polr()` or `ordinal::clm()` using the same ordered response, rows, formula and logit link. The coefficient vectors agree to numerical tolerance (within 10^{-5} here), and both give AIC 10420.5924. Different function defaults or reversed category order would prevent this direct comparison.

```
bridge <- MASS::polr(poverty ~ religion + degree + country + age + gender,
                     data = opinion, Hess = TRUE)
AIC(bridge, po)
nominal_test(po)
```

Relaxing the country restriction allows a separate country comparison at each cut point, while retaining common slopes for religion, degree, age and gender. The fitted signs must be translated to the same above-cut direction before comparison. Table 18 reports that direction: a positive coefficient increases the log-odds of a response **above** the named split, conditional on the remaining variables. Australia is the reference country.

Table 18

Country comparisons at the two cumulative splits, with common slopes for the remaining recorded predictors.

Comparison	Above-cut coefficient	SE	Odds ratio
Norway, lower split	-0.1227	0.0778	0.8845
Norway, upper split	-1.7812	0.1817	0.1684
Sweden, lower split	-0.4449	0.0824	0.6409
Sweden, upper split	-2.0687	0.2134	0.1264
USA, lower split	0.3626	0.0734	1.4370
USA, upper split	0.8727	0.0867	2.3935

The remaining coefficients and thresholds complete the partial model. Thresholds retain the cumulative-link convention $\zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta}$; they receive no odds-ratio interpretation.

Table 19*Thresholds and common slopes in the partial proportional-odds fit.*

Parameter	Estimate	SE	Odds ratio
Lower threshold	0.7178	0.1040	—
Upper threshold	2.3609	0.1146	—
Religion: yes versus no	0.1491	0.0762	1.1608
Degree: yes versus no	0.1414	0.0666	1.1519
Age per year	0.0106	0.0016	1.0107
Male versus female	0.1738	0.0529	1.1899

The likelihood-ratio comparison with the common-slope model is nested: it frees three country contrasts at a second cut point. It gives $\chi^2(3) = 362.3463$, $p < .0001$, and reduces AIC from 10420.5924 to 10064.2461. This is evidence against a common country slope in this specified model. Other restrictions and the observational interpretation still require judgment. Allowing country to vary by cut does not establish that every other term satisfies proportional odds.

```
anova(po, cutfit)
AIC(po, cutfit)
```

The probability comparison below holds religion at “yes”, university degree at “no”, age at 45 and gender at “female”. The profiles define what is being compared. They are not national population averages. Probabilities are reported in the recorded order Too Little, About Right, Too Much; inspect the changed allocation across all three, rather than treating a threshold coefficient as a category probability.

Table 20*Profile probabilities under common and country-specific cumulative slopes.*

Country	Model	Too Little	About Right	Too Much
Australia	Common slopes	0.5122	0.3521	0.1357
Norway	Common slopes	0.5917	0.3062	0.1021
Sweden	Common slopes	0.6575	0.2634	0.0791
USA	Common slopes	0.3614	0.4130	0.2255
Australia	Cut-specific country	0.5229	0.3271	0.1500
Norway	Cut-specific country	0.5533	0.4178	0.0289
Sweden	Cut-specific country	0.6310	0.3472	0.0218
USA	Cut-specific country	0.4326	0.2704	0.2970

An adjacent-category logit instead compares neighbouring categories; a continuation-ratio logit describes successive transitions conditional on reaching a stage. Those models answer differently

structured questions. Category order is the starting point, but the response process and the desired comparison determine which ordered model is useful (Agresti, 2010).

Appendix B: Residuals and Two-process Explanations

Residuals and one influential record

A Pearson residual divides the observed-minus-fitted difference by the model's fitted standard deviation. A deviance residual gives a signed contribution to discrepancy under the likelihood. Neither should be read as an ordinary Normal residual merely because it is called a residual. Discreteness and changing means shape both displays.

The largest absence count is 81 days. Removing that record asks a sensitivity question; it does not establish that the record is an error. Refitting without it should report the changed dispersion parameter, the ethnicity interval and the continued need for a plausible variance model. Automatic deletion would conceal the very upper-tail behaviour the model is meant to explain.

```
reduced <- subset(absence, Days < 81)
check <- MASS::glm.nb(Days ~ Eth + Sex + Age + Lrn, data = reduced)
check$theta
exp(confint.default(check)["EthN", ])
```

When zero has a separate mechanism

A pupil who was eligible to miss school but happened to attend every day and a record that could never register an absence can both show zero. The first belongs to the same count process as other pupils. The second suggests a separate eligibility or recording process. Many zeros alone do not distinguish them: an ordinary negative-binomial model can already produce more zeros than a Poisson model with the same mean.

A **hurdle model** separates whether the count crosses zero from its positive magnitude; its positive-count component is truncated at zero. A **zero-inflated model** mixes a structural-zero component with an ordinary count component that can itself produce zero (Lambert, 1992; Mullahy, 1986). These are different accounts of how records arise. An extra zero parameter should be motivated by that account and checked against its fitted consequences.

Comparisons with an ordinary count model can put mixture probabilities on a boundary and can leave additional parameters unidentified under the null. The usual interior chi-square likelihood-ratio reference is therefore not automatic, nor is one universal half-and-half mixture formula adequate for every such comparison. Specialist model-specific methods or a carefully designed parametric bootstrap are needed before attaching a calibrated p-value.

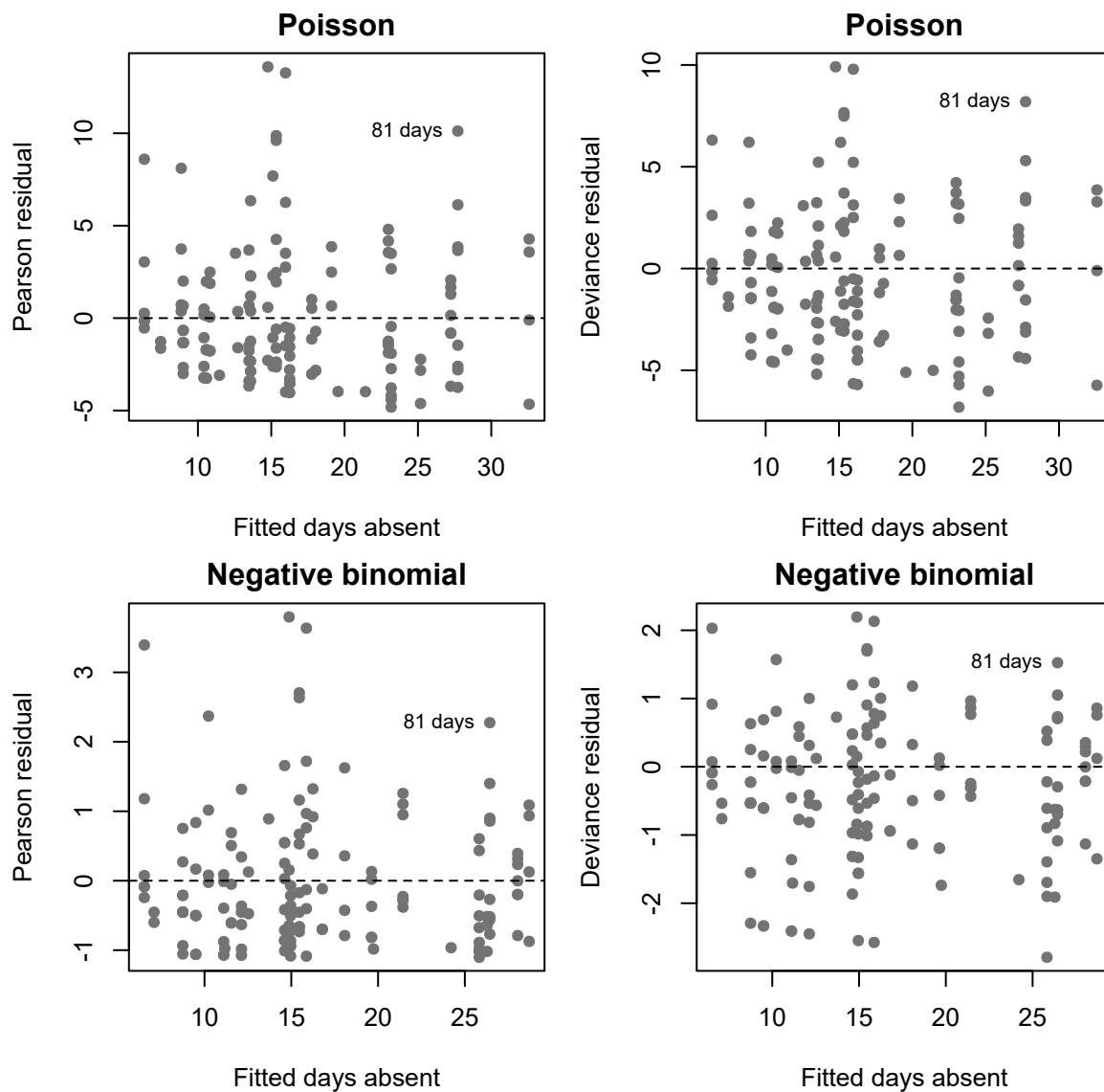


Figure 13: Residuals on each count model's own variance and likelihood scales.

For further practice, explain why the quasi-Poisson fit cannot supply the band-probability table, and carry out the 81-day sensitivity comparison. In each case, distinguish what changes numerically from what still needs a study-design or measurement justification.

References

- Agresti, A. (2010). *Analysis of ordinal categorical data* (2nd ed.). Wiley. <https://doi.org/10.1002/9780470594001>
- Agresti, A. (2018). *Statistical methods for the social sciences* (5th ed.). Pearson.
- Cameron, A. C., & Trivedi, P. K. (2013). *Regression analysis of count data* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/CBO9781139013567>
- Christensen, R. H. B. (2025). *ordinal: Regression models for ordinal data* (Versions 2025.12-29). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.ordinal>
- Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. <https://www.john-fox.ca/Companion/index.html>
- Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.carData>
- Hilbe, J. M. (2011). *Negative binomial regression* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/CBO9780511973420>
- Lambert, D. (1992). Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics*, 34(1), 1–14. <https://doi.org/10.2307/1269547>
- Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- McCullagh, P. (1980). Regression models for ordinal data. *Journal of the Royal Statistical Society: Series B (Methodological)*, 42(2), 109–142. <https://doi.org/10.1111/j.2517-6161.1980.tb01109.x>
- Mullahy, J. (1986). Specification and testing of some modified count data models. *Journal of Econometrics*, 33(3), 341–365. [https://doi.org/10.1016/0304-4076\(86\)90002-3](https://doi.org/10.1016/0304-4076(86)90002-3)
- Nelder, J. A., & Wedderburn, R. W. M. (1972). Generalized linear models. *Journal of the Royal Statistical Society: Series A (General)*, 135(3), 370–384. <https://doi.org/10.2307/2344614>
- Venables, W. N., & Ripley, B. D. (2002). *Modern applied statistics with S* (4th ed.). Springer. <https://doi.org/10.1007/978-0-387-21706-2>
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2), 129–133. <https://doi.org/10.1080/00031305.2016.1154108>